

MIMO Channel Memory Requirements for Randomly-Coupled Multicore Fiber Networks

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Abstract We evaluate the impact of network nodes on MIMO memory requirements in a C+L-band randomly-coupled multicore fiber network. While point-to-point transmission requires less than 1 ns of memory, constraining network operation to this length results in throughput penalties of 50–75% due to path delay misalignment. © 2026 The Author(s)

Introduction

Space-division multiplexing (SDM) is a promising solution to the capacity crunch in optical fiber networks [1–3]. Spatial super-channel (SSC) switching networks supported by high-spatial density randomly-coupled multicore fibers (RC-MCFs) have the potential to provide the necessary capacity [4, 5]. Furthermore, the use of shared switching resources, such as liquid crystal on silicon (LCOS) devices, over all spatial channels (SCs) significantly reduces the overall required switching fabric [6–10]. Nevertheless, data transmission over RC-MCFs is conditioned by physical-layer impairments, particularly SC coupling and group delay spread. These can be handled using multiple-input multiple-output (MIMO) subsystems, with increasing complexity with the number of SCs and the group delay spread [11]. Even more relevant to SSC networks are inter-SC delays introduced by network elements, shown to rapidly exceed the delays from fiber propagation [8, 9].

Here, we experimentally investigate the impact of network switching nodes on MIMO memory lengths in a C+L-band RC-MCF network test-bed. We consider a multi-band SSC switching node operating under high-spatial channel counts and long transmission distances. To this end, we implement a 2-line side network element supporting

386 SSCs across the C- and L-bands, and evaluate network scenarios using 19-core, 63 km and 86 km length RC-MCFs. We compare network operation with point-to-point (PTP) transmission over the same fibers to assess the impact of the node on MIMO memory requirements. The results show that, while PTP transmission requires sub-nanosecond memory lengths, network operation increases the required memory to 3.3 ns due to path-dependent temporal misalignment. Constraining network operation to transmission memory lengths leads to severe throughput penalties of 50-75%, indicating limited tolerance to node path misalignment without increasing MIMO memory lengths. This work highlights the impact and importance of path alignment in the design of SSC switching nodes for RC-MCF networks.

Experimental Setup

Fig. 1-a) shows a simplified diagram of the implemented 2-line side network element. We considered inputs from the add ports and the west line side, and outputs from the drop ports and the east line side. We used a 384×384 wavelength-agnostic switch to direct the signals. Each line side consisted of $19 \times (1 \times 2)$ wavelength cross-connects (WXC), implemented with specially designed multi-port wavelength selective switches

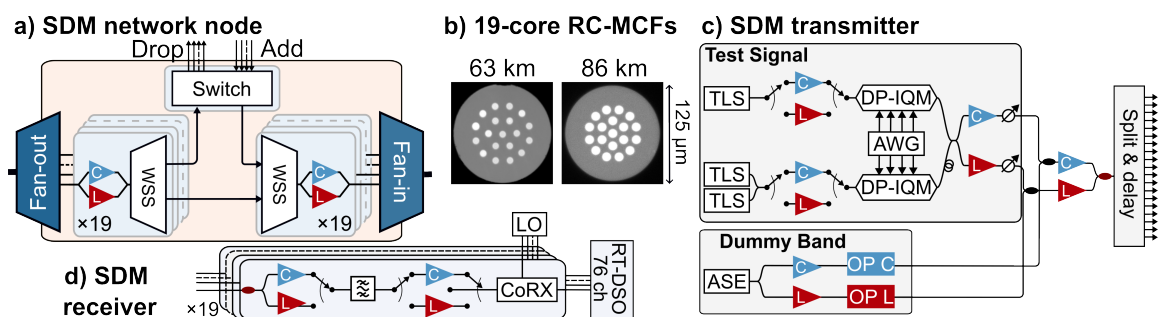


Fig. 1: Simplified diagram of the experimental setup of: a) C+L-band 2-line side network node, b) profiles of the 63 km and 86 km 19-core randomly-coupled multicore fibers, c) SDM transmitter and d) SDM receiver.

(WSSs) that used a single LCOS device to simultaneously switch 7 SCs. This enabled SSC switching of 19 SCs using only 3 LCOS devices, reducing switching resources and path misalignment. The 2-line sides were amplified using band-specific erbium-doped fiber amplifiers (EDFAs) to compensate for WSS loss. The relative delays among the spatial paths within each set of 19 EDFAs and WSSs were aligned within 1.5 ns, using single-mode fiber patch cords. Each line side was connected to fan-in and fan-out devices to couple the SCs into and out of the two RC-MCFs, respectively. The used 19-core RC-MCFs, shown in Fig. 1-b), were 63 km and 86 km long. The two fibers were drawn from different preforms and had slightly different properties, described in [12].

Fig. 1-c) shows the experimental setup of the SDM transmitter. A 24.5 GBaud polarization-multiplexed (PM)-16-QAM signal was generated by modulating the output of 3 tunable laser sources (TLSs) using dual-polarization IQ modulators (DP-IQMs), driven by an arbitrary-waveform generator (AWG) operating at 245 GSa/s. A dummy band was produced by amplifying and spectrally shaping amplified spontaneous emission (ASE) noise using optical processors (OPs), which also carved a 75 GHz notch to accommodate the test band. The combined test and dummy band formed the equivalent to a full wavelength-division multiplexing signal, with 386 25 GHz-spaced channels spanning the C- and L-bands. The resulting signal was split into 19 replicas, each delayed by 50 ns for decorrelation to emulate 19 independent SCs.

Fig. 1-d) shows the SDM receiver. The 19 SCs

were received with 19 parallel receivers, each consisting of a two-stage amplification, with an optical bandpass filter (BPF) in between to isolate the wavelength channel under test. The signals were mixed with a common local oscillator (LO) and detected using 19 coherent receivers. The resulting signals were digitized with a 76-channel real-time digital storage oscilloscope (RT-DSO) and stored for offline signal processing. The digital signal processing (DSP) operated at 2 samples per symbol and included chromatic dispersion compensation and a 38×38 time-domain MIMO equalization to undo the spatial coupling and compensate for the channel impulse response. The MIMO taps were estimated using a data-aided least mean square algorithm. Carrier recovery was performed within the equalizer loop [13].

Results and Discussion

Fig. 2-a) and b) show diagrams of PTP transmission over 63 km and 86 km 19-core RC-MCFs and the evaluated network scenarios. Scenarios 1 and 2 corresponded to full add&drop of 386 SSCs over the 63 km and 86 km RC-MCFs, respectively. Scenario 3 represented full express operation, while scenario 4 involved partial add&drop of 7 SSCs in the center of each band, with optical grooming of 372 optically bypassed SSCs. Fig. 2-c) shows the MIMO intensity impulse response (IIR) in PTP transmission, calculated as the sum of the squared magnitudes of the equalizers [14]. The memory length was estimated from the time interval containing 98% of the energy in the IIR, with a 10% margin. The RC-MCFs had similar

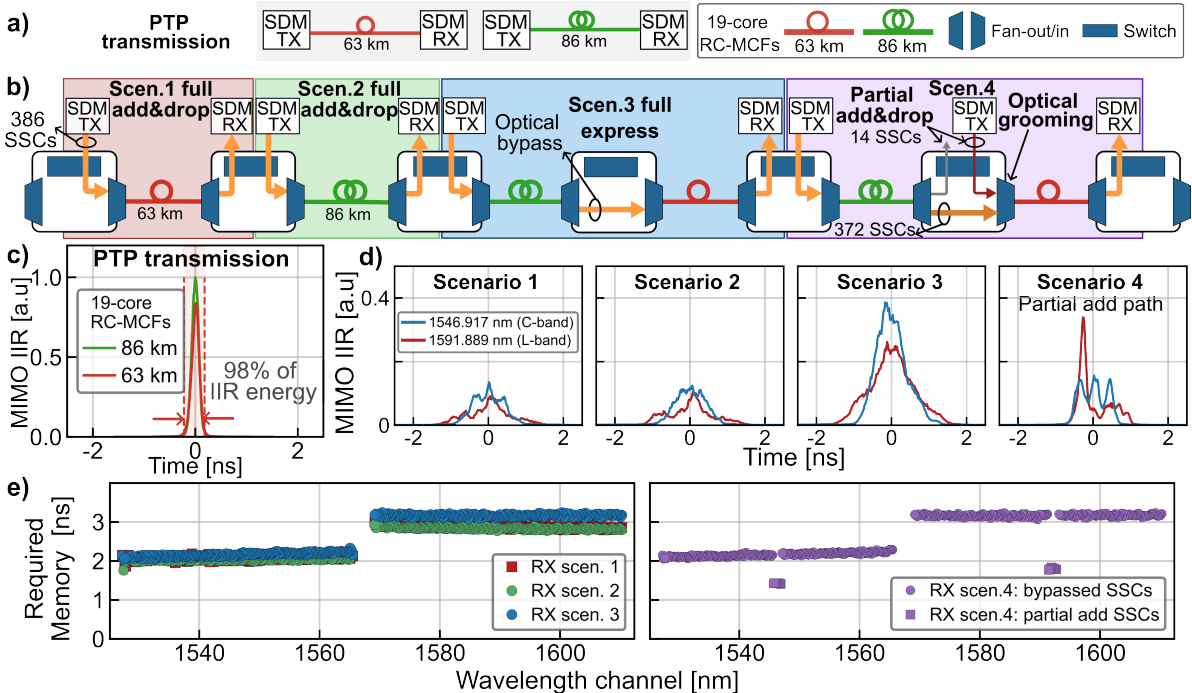


Fig. 2: Simplified diagram of a) PTP transmission over 63 km and 86 km 19-core RC-MCFs and b) evaluated network scenarios. MIMO IIRs in c) PTP transmission and d) network scenarios at 1546.917 nm and 1591.889 nm, corresponding to C- and L-band SSCs, respectively. e) Required memory to cover 98% of the area under the IIR, shown for each SSC in the network scenarios.

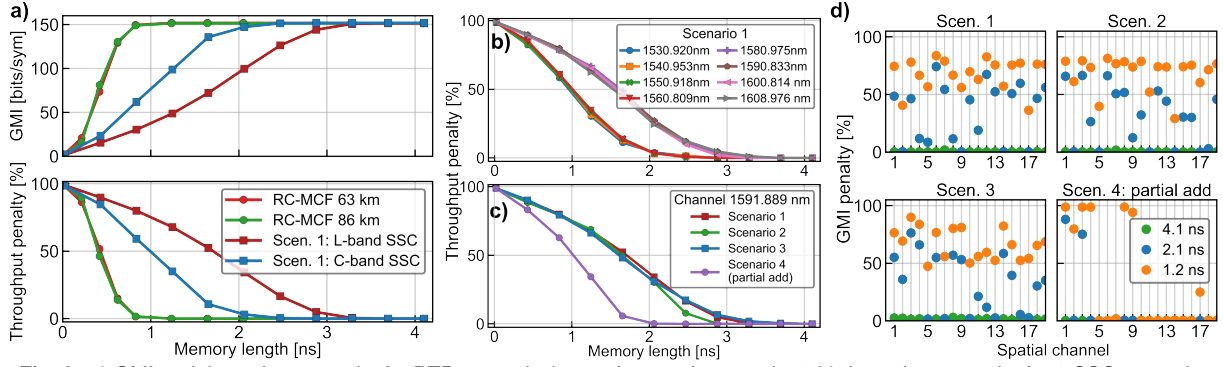


Fig. 3: a) GMI and throughput penalty for PTP transmission and network scenario 1, b) throughput penalty for 8 SSCs over the supported wavelengths in scenario 1, c) L-band SSC in network scenarios, and d) per-SC GMI penalty for an L-band SSC.

Gaussian-shaped IIR, indicating random coupling among the different SCs, with memory lengths around 400 ps. Fig. 2-d) shows the IIR for a center C- and L-band SSC in the network scenarios. Compared to PTP transmission, the IIRs were significantly broader. The IIR shape depended on the routing path and differed between C- and L-band SSCs, with distinct peaks indicating temporal delays between SCs. Fig. 2-e) shows the memory lengths for all SSCs. Scenarios 1 and 2 differed only in transmission fiber, resulting in similar memory lengths with an average of 2.1 ns in the C-band and 2.8 ns in the L-band. Scenario 3 required slightly larger memory due to propagation over two fiber spans, averaging to 2.2 ns and 3.2 ns in the C- and L-bands, respectively. In scenario 4, memory lengths ranged from 1.5 to 3.3 ns. Partially added SSCs required the shortest memory, as they traversed node elements only once. The results showed that memory length was dominated by path delays introduced by node elements.

Fig. 3 summarizes the impact of the network node on the required MIMO memory length. Performance was evaluated using generalized mutual information (GMI) [15], and throughput penalty relative to maximum PTP GMI. Fig. 3-a) compares PTP transmission and scenario 1. In PTP transmission, a maximum GMI of approximately 152 bit/symbol was achieved with less than 1 ns memory for both fibers. Applying this memory in scenario 1 resulted in throughput penalties of 53% and 75% for the C- and L-band SSCs, respectively. Matching PTP performance in the network required memory lengths of 2.4 ns (C-band SSC) and 3.3 ns (L-band SSC), consistent with the IIR duration. Shorter memory did not fully compensate for spatial mixing, leading to performance degradation. Fig. 3-b) shows similar penalties across channels within each band but higher penalties for L-band SSCs. Fig. 3-c) shows that the penalty depended on the routing path. The partial add path required the shortest memory, but still incurred a 50% penalty when using PTP transmission memory length. Scenarios 1 to 3 showed comparable

behavior. For a fixed-memory MIMO subsystem, the required memory to avoid penalty across all scenarios was determined by the longest path delay, estimated here to be around 3.3 ns.

Fig. 3-d) shows the GMI penalty per SC relative to PTP transmission for an L-band SSC. With sufficient memory, no penalty was observed, indicating full compensation of spatial mixing. Reducing memory introduced penalties in a subset of SCs, likely associated with a larger temporal spread. Scenarios in which the IIR was closer to a Gaussian shape showed penalties for most SCs, while scenarios with clearly distinct peaks in the IIR, such as the partial add case, showed higher penalties confined to specific SCs, with remaining SCs unaffected.

In the considered network, transmission IIRs were around 400 ps, while node path delays were aligned within 1.5 ns. As a result, MIMO memory requirements were dominated by node delay misalignment rather than fiber propagation. The larger penalties observed in the L-band indicated that most of the mismatch originated from different paths through band-specific EDFAs. These results highlight that aligning node paths within the transmission IIR duration is a critical design constraint to limit MIMO DSP complexity, as increasing delay mismatch directly increases memory demands.

Conclusion

We experimentally showed that MIMO memory requirements in C+L-band SDM networks over 19-core RC-MCFs are strongly impacted by network node delay misalignment. While PTP transmission required sub-ns memory, with fiber IIRs around 400 ps, node path delays increased the required memory in network operation up to 3.3 ns, depending on the SSC routing path. Using the same memory length as in PTP transmission resulted in severe throughput penalties between 50% and 75%. This shows that constraining node path delays to the transmission IIR duration is essential in a node design to reduce MIMO memory requirements and maintain the benefits of RC-MCFs in SSC switching-based networks.

Acknowledgements

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